SECTION A: Scenario Based Conceptual Questions

1. An online learning platform releases new courses every month. Identify the most suitable software development life cycle model and justify your choice

Recommended Model: Agile

Justification:

- An online learning platform releases new courses every month — this requires short, iterative development cycles which Agile supports perfectly.
- Agile allows incremental delivery, so each sprint (2-4 weeks) can release a set of new courses or features without waiting for a full product cycle.
- Continuous feedback from learners and instructors can be incorporated quickly into the next sprint.
- The Agile model accommodates changing requirements (new course formats, updated curricula) without disrupting the entire project.
- It supports parallel workstreams — developers, content creators, and testers can work concurrently.

Why NOT Waterfall: Waterfall requires fully defined requirements upfront and does not accommodate monthly changes easily.

Why NOT V-Model: Although good for testing, it is sequential and rigid — unsuitable for a dynamic, fast-release platform.

2. For a course purchase feature, define testing objectives from a quality assurance perspective.

Testing Objectives:

- Verify that users can successfully search, select, and purchase a course.
- Ensure payment gateway integration functions correctly for all payment methods (credit/debit card, UPI, wallet, net banking).
- Validate that payment failure scenarios show appropriate error messages and do not deduct money.
- Confirm that purchase confirmation emails/notifications are sent correctly after successful transactions.
- Ensure course access is granted immediately after successful payment.
- Test refund and cancellation workflows work as per business rules.

- Validate security: SSL encryption, PCI-DSS compliance, and no data leakage during transactions.
- Performance: Ensure the payment flow handles concurrent users without timeout or crashes.
- Verify correct invoice/receipt generation with all pricing, tax, and discount details.
- Cross-browser and cross-device compatibility for the purchase flow.

3. Explain how non-functional testing contributes to better learner experience.

Non-functional testing focuses on HOW the system performs rather than WHAT it does. It directly impacts learner satisfaction:

NFT Type	Impact on Learner Experience
Performance Testing	Ensures videos load fast, pages respond quickly — no buffering during lessons
Load Testing	Validates platform stability when thousands of learners access simultaneously (e.g., during course launch)
Usability Testing	Confirms the UI/UX is intuitive — learners can navigate courses easily without confusion
Security Testing	Protects learner data, payment info, and prevents unauthorized access
Compatibility Testing	Ensures courses work across browsers (Chrome, Firefox, Safari) and devices (mobile, tablet, desktop)
Reliability Testing	Validates uptime — learners can access courses 24/7 without unexpected downtime
Scalability Testing	Ensures platform grows with user base without degradation in speed or quality
Accessibility Testing	Ensures learners with disabilities can use screen readers, keyboard navigation, etc.

4. During sprint execution, how should testers ensure acceptance criteria are testable?

Steps testers should follow:

- Use the SMART Criteria: Ensure each acceptance criterion is Specific, Measurable, Achievable, Relevant, and Time-bound.

- Participate in Sprint Planning: Testers must attend sprint planning meetings and collaborate with Product Owners to refine user stories.
- Apply the Given-When-Then (GWT) format: Rewrite acceptance criteria in a structured, testable format. Example — Given a registered user, When they log in with valid credentials, Then they should be redirected to the dashboard.
- Identify ambiguous criteria early: Flag vague statements like 'the system should be fast' and replace with measurable ones like 'page load time should be under 3 seconds'.
- Create test scenarios directly from acceptance criteria: Each criterion should map to at least one test case.
- Use Definition of Ready (DoR): Only accept user stories into the sprint if acceptance criteria are clear and testable.
- Conduct Three Amigos meetings: Developer + Tester + Product Owner discuss each story to align understanding before development begins.
- Maintain traceability: Link each test case to its acceptance criterion in the RTM.

5. A payment failure occurs after deployment. Assign severity and priority with justification.

Defect: Payment failure occurs after deployment on the live application.

Severity: CRITICAL

Justification for Severity: The payment module is a core business function. Failure means users cannot purchase courses — the primary revenue-generating feature of the platform is completely broken. Data integrity is at risk (partial transactions, incorrect balances).

Priority: P1 — Highest Priority (Immediate Fix Required)

Justification for Priority: This is a production issue affecting live users and causing direct revenue loss. Business impact is severe — every minute of downtime results in financial loss and reputational damage. An immediate hotfix, rollback, or patch must be deployed. All other tasks must be deprioritized until this is resolved.

Attribute	Value	Reason
Severity	Critical	Core feature completely broken in production
Priority	P1 (Highest)	Direct revenue impact, affects all purchasing users
Resolution Time	Immediate (<2 hours)	Live production environment, customer-facing
Action	Hotfix / Rollback	Cannot wait for next sprint

SECTION B: Practical Manual Testing Tasks

1. STLC — Student Login Module

Phase 1: Requirement Analysis

Activities:

- Review functional requirements for Student Login (email/password, social login, forgot password).
- Identify login-related user stories and acceptance criteria from the product backlog.
- Clarify doubts with the BA/PO (e.g., session timeout duration, account lockout policy).
- Identify non-functional requirements — response time, security (HTTPS, encryption).

Deliverable: Requirement Understanding Document / Test Basis

Phase 2: Test Planning

Activities:

- Define testing scope: What to test (valid/invalid login, session management, forgot password) and what not to test (backend DB logic).
- Estimate effort and timeline for test design, execution, and reporting.
- Identify test environment: Browser versions, OS, device types.
- Define test data requirements: Valid/invalid credentials, locked accounts, unregistered emails.
- Choose testing types: Functional, Boundary, Negative, Security, Usability.

Deliverable: Test Plan Document

Entry Criteria for Student Login Testing

- Login module development is complete and deployed to test environment.
- Requirement documents and acceptance criteria are available and approved.
- Test environment is set up and accessible.
- Test data (valid/invalid credentials) is prepared.
- All critical dependencies (database, email server) are available.

Exit Criteria for Student Login Testing

- All planned test cases have been executed.

- 95% or more test cases have passed.
- All Critical and High severity defects are resolved and verified.
- No open P1/P2 defects related to login functionality.
- Test Summary Report is signed off by the QA Lead and Product Owner.

Phase 3: Test Design

TC ID	Test Scenario	Type
TC_LG_001	Login with valid email and correct password	Positive
TC_LG_002	Login with valid email and wrong password	Negative
TC_LG_003	Login with unregistered email	Negative
TC_LG_004	Login with empty email and password fields	Negative
TC_LG_005	Login with correct email but empty password	Negative
TC_LG_006	Verify account lockout after 5 failed attempts	Security
TC_LG_007	Verify Forgot Password link works correctly	Functional
TC_LG_008	Verify Remember Me functionality persists session	Functional
TC_LG_009	Verify login via Google OAuth	Integration
TC_LG_010	Verify session expires after timeout period	Security

Phase 4: Test Execution

- Execute all test cases against the Student Login module in the test environment.
- Log defects with screenshots, steps to reproduce, severity, and priority.
- Perform retesting after defect fixes.
- Perform regression testing to ensure existing functionality is not broken.

Deliverable: Executed Test Cases with Pass/Fail status, Defect Reports

Phase 5: Test Closure

- Prepare Test Summary Report with overall execution results.
- Document lessons learned and improvement suggestions.
- Archive all test artifacts (test cases, defect logs, reports).

- Obtain sign-off from stakeholders.

Deliverable: Test Closure Report

2. Test Design — Course Rating (1–5 Stars)

Equivalence Partitioning (EP)

Partition	Range	Description	Type
Valid Partition	1 to 5	All integer values from 1 to 5	Valid
Invalid Partition 1	Less than 1 (e.g., 0, -1, -5)	Values below minimum	Invalid
Invalid Partition 2	Greater than 5 (e.g., 6, 10, 100)	Values above maximum	Invalid
Invalid Partition 3	Decimal values (e.g., 2.5, 3.7)	Non-integer values	Invalid
Invalid Partition 4	Alphabetic / Special chars (e.g., 'A', '@')	Non-numeric input	Invalid

Boundary Value Analysis (BVA)

Test Value	Type	Expected Result
0	Just below minimum (Invalid)	Error: Rating must be between 1 and 5
1	Minimum valid boundary	Rating accepted — 1 star
2	Just above minimum (Valid)	Rating accepted — 2 stars
3	Middle valid value	Rating accepted — 3 stars
4	Just below maximum (Valid)	Rating accepted — 4 stars
5	Maximum valid boundary	Rating accepted — 5 stars
6	Just above maximum (Invalid)	Error: Rating must be between 1 and 5
-1	Negative value (Invalid)	Error: Invalid rating input
2.5	Decimal value (Invalid)	Error: Only whole numbers allowed
ABC	Non-numeric (Invalid)	Error: Please enter a valid number

3. Test Cases — Course Experience

Test Case 1: Course Search & Browse

Field	Details

Test Case ID	TC_CS_001
Test Case Title	Verify course search returns relevant results
Precondition	User is logged in. At least 3 courses exist in the system.
Test Steps	1. Navigate to the homepage. 2. Click on the Search bar. 3. Type 'Python Programming'. 4. Press Enter or click Search icon. 5. Observe search results page.
Expected Result	System displays a list of courses matching 'Python Programming' with course name, instructor, rating, and price. Results are sorted by relevance.
Actual Result	Courses related to Python Programming are displayed correctly.
Pass/Fail Status	PASS

Test Case 2: Course Learning Experience

Field	Details
Test Case ID	TC_CL_001
Test Case Title	Verify enrolled student can play course video lectures
Precondition	User is logged in and has purchased/enrolled in the course.
Test Steps	1. Navigate to 'My Courses'. 2. Click on an enrolled course. 3. Click on Lecture 1 in the curriculum. 4. Click the Play button. 5. Observe video playback.
Expected Result	Video plays smoothly without buffering. Progress bar is visible. Student can pause, resume, rewind, and adjust volume/quality.
Actual Result	Video plays correctly with all playback controls functioning.
Pass/Fail Status	PASS

Test Case 3: Assessments & Quizzes

Field	Details
Test Case ID	TC_AQ_001

Test Case Title	Verify student can attempt and submit a quiz
Precondition	User is enrolled in a course that has a quiz. Quiz has 5 MCQ questions.
Test Steps	1. Open the enrolled course. 2. Navigate to the Quiz section. 3. Click 'Start Quiz'. 4. Answer all 5 questions. 5. Click 'Submit Quiz'. 6. Observe result screen.
Expected Result	Quiz is submitted successfully. Score (e.g., 4/5 = 80%) is displayed with correct/incorrect answers highlighted. Certificate or completion badge is awarded if score meets passing threshold.
Actual Result	Quiz submitted and score displayed correctly with detailed feedback.
Pass/Fail Status	PASS

4. Defect Log — 3 Unique Defects

Defect 1: Invalid Email Accepted During Login

Field	Details
Defect ID	DEF_001
Defect Title	System accepts invalid email format during Student Login
Severity	High
Priority	P2
Module	Student Login
Steps to Reproduce	1. Go to Login page. 2. Enter 'testuser' (no @ symbol) in email field. 3. Enter any password. 4. Click Login.
Expected Result	System should show validation error: 'Please enter a valid email address'
Actual Result	System proceeds to authenticate without email format validation, showing a server error
Environment	Browser: Chrome 124, OS: Windows 11, Environment: QA/Test

Defect 2: Course Rating Accepts Value Above 5

Field	Details
Defect ID	DEF_002

Defect Title	Course rating input field accepts values greater than 5
Severity	Medium
Priority	P2
Module	Course Rating
Steps to Reproduce	1. Log in and go to a completed course. 2. Click 'Rate this Course'. 3. Manually type '8' in the rating input field. 4. Click Submit.
Expected Result	System should reject the input and show error: 'Rating must be between 1 and 5'
Actual Result	Rating of 8 is accepted and saved. Average course rating becomes incorrect.
Environment	Browser: Firefox 125, OS: macOS Ventura, Environment: QA/Test

Defect 3: Quiz Timer Does Not Stop After Submission

Field	Details
Defect ID	DEF_003
Defect Title	Quiz countdown timer continues running after quiz is submitted
Severity	Medium
Priority	P3
Module	Assessments & Quizzes
Steps to Reproduce	1. Log in and open a course with a timed quiz (10-minute timer). 2. Click 'Start Quiz'. 3. Answer all questions within 3 minutes. 4. Click 'Submit Quiz'. 5. Observe the result page.
Expected Result	Timer should stop immediately upon submission. Result page should show time taken.
Actual Result	Timer continues to count down on the result page even after submission. UI appears broken.
Environment	Browser: Safari 17, OS: iOS 17 (iPad), Environment: QA/Test

SECTION C: Mini Capstone Project — End-to-End Testing Case Study

1. Test Planning Document

1. Project Overview

Project: Online Learning Platform — End-to-End Testing

Version: 1.0 | Prepared By: QA Team | Date: June 2026

2. Scope

In Scope:

- Student Registration and Login
- Course Search, Browse, and Selection
- Course Purchase and Payment Processing
- Course Access, Video Playback, and Progress Tracking
- Assessments and Quizzes
- Course Rating and Review

Out of Scope:

- Instructor-side portal (course creation, analytics)
- Admin dashboard
- Third-party payment gateway internal logic

3. Objectives

- Validate all critical user flows function correctly from login to course completion.
- Identify and report defects before production release.
- Ensure the platform meets performance and security standards.
- Verify all acceptance criteria defined in user stories are met.

4. Testing Types

Testing Type	Purpose
Functional Testing	Verify each feature works as per requirements
Integration Testing	Validate interaction between Login, Payment, and Course Access modules
Regression Testing	Ensure new changes do not break existing functionality

Performance Testing	Validate response times under normal and peak load
Security Testing	Test for XSS, SQL Injection, unauthorized access vulnerabilities
UAT (User Acceptance Testing)	Business sign-off before go-live
Compatibility Testing	Cross-browser and cross-device validation

5. Test Environment

- OS: Windows 11, macOS Ventura, iOS 17, Android 14
- Browsers: Chrome 124, Firefox 125, Safari 17, Edge 124
- Test URL: https://test.learningplatform.com
- Database: PostgreSQL 15 (Test instance)
- Payment Gateway: Stripe Test Mode

6. Risks & Mitigations

Risk	Mitigation
Payment gateway unavailability	Use sandbox/test mode throughout QA cycle
Incomplete requirements	Conduct Three Amigos meetings; get BA sign-off before sprint start
Test data unavailability	Prepare test data scripts; use data masking for sensitive info
Resource unavailability	Cross-train team members; maintain backup testers
Environment instability	Set up dedicated stable QA environment separate from Dev

7. Assumptions

- All functional requirements are finalized and approved by stakeholders.
- Test environment will be available throughout the testing cycle.
- Developers will fix Critical/High defects within 24 hours of reporting.
- Stripe payment gateway sandbox is configured and operational.

8. Entry and Exit Criteria

Entry Criteria:

- All modules are code-complete and deployed to the test environment.
- Requirement documents and user stories with acceptance criteria are available.
- Test data is prepared and test environment is stable.

Exit Criteria:

- All planned test cases are executed.
- 95%+ test cases passed.
- Zero open Critical/P1 defects.
- Test Summary Report reviewed and signed off by QA Lead and PO.

2. Requirements Traceability Matrix (RTM)

Req ID	Requirement	Test Scenario	Test Case ID	Status	Defect ID
REQ_001	Student Login with valid credentials	Verify successful login	TC_LG_001	PASS	-
REQ_002	Login validation for invalid credentials	Verify error on wrong password	TC_LG_002	PASS	-
REQ_003	Email format validation on login	Verify invalid email rejected	TC_LG_003	FAIL	DEF_001
REQ_004	Course search by keyword	Verify search results relevance	TC_CS_001	PASS	-
REQ_005	Course purchase via payment gateway	Verify end-to-end purchase flow	TC_PY_001	PASS	-
REQ_006	Course access after payment	Verify enrolled course playback	TC_CL_001	PASS	-
REQ_007	Quiz attempt and submission	Verify quiz submit and score	TC_AQ_001	PASS	-
REQ_008	Course rating 1–5 validation	Verify boundary values rejected	TC_RT_001	FAIL	DEF_002
REQ_009	Quiz timer functionality	Verify timer stops on submission	TC_AQ_002	FAIL	DEF_003

3. End-to-End Test Scenario: Login → Course Selection → Payment → Course Access

Scenario ID: E2E_001

Field	Details
Scenario Title	Complete learner journey from Login to Course Access
Objective	Verify that a registered student can log in, find a course, pay for it, and access the course content successfully
Preconditions	1. Test user account exists: student@test.com / Test@1234. 2. 'Python for Beginners' course is published and priced at Rs. 999. 3. Payment gateway (Stripe) is in sandbox mode. 4. User does NOT already own this course.

Detailed Steps:

Step	Action	Expected Result	Status
1	Navigate to https://test.learningplatform.com	Homepage loads within 3 seconds	PASS
2	Click 'Login' button on the top navigation bar	Login page is displayed with email, password fields and social login options	PASS
3	Enter email: student@test.com, Password: Test@1234	Credentials entered in respective fields	PASS
4	Click 'Login' button	User is authenticated and redirected to the dashboard. Welcome message displayed.	PASS
5	Click the Search bar and type 'Python for Beginners'	Autocomplete suggestions appear	PASS
6	Press Enter to search	Search results page shows relevant Python courses with name, rating, price, instructor	PASS

7	Click on 'Python for Beginners' course	Course detail page loads with curriculum, reviews, instructor info, and 'Enroll Now' button	PASS
8	Click 'Enroll Now / Buy Now' (Price: Rs. 999)	User is redirected to the order summary / checkout page	PASS
9	Select payment method: Credit Card. Enter test card: 4111 1111 1111 1111, Exp: 12/28, CVV: 123	Payment details entered in secure form (HTTPS)	PASS
10	Click 'Pay Now'	Payment is processed. Stripe sandbox returns success. Order confirmation page shown with invoice.	PASS
11	Check email inbox for test account	Order confirmation email received with course details and invoice PDF	PASS
12	Navigate to 'My Courses'	Purchased course 'Python for Beginners' appears in My Courses	PASS
13	Click on the course	Course curriculum is displayed with all lectures and sections unlocked	PASS
14	Click on Lecture 1 and press Play	Video plays smoothly. Progress bar advances. Controls (pause, seek, volume) work correctly.	PASS

15	Complete Lecture 1. Check progress bar.	Progress shows 'Lecture 1 Complete'. Completion percentage updates (e.g., 5%)	PASS

Overall Result: PASS — Complete E2E flow verified successfully.

4. Test Summary Report

1. Executive Summary

The End-to-End testing of the Online Learning Platform has been completed. Testing covered Student Login, Course Search, Course Purchase, Course Access, Quizzes, and Course Rating modules. Overall, the platform is functionally stable with minor defects identified and documented.

2. Test Execution Status

Metric	Count
Total Test Cases Planned	25
Test Cases Executed	25
Test Cases Passed	22
Test Cases Failed	3
Test Cases Blocked	0
Pass Rate	88%

3. Defects Summary

Defect ID	Title	Severity	Priority	Status
DEF_001	Invalid email format accepted at login	High	P2	Open — Fix in Progress
DEF_002	Course rating accepts values above 5	Medium	P2	Open
DEF_003	Quiz timer continues after submission	Medium	P3	Open

4. Defect Distribution by Severity

Severity	Count
Critical	0
High	1
Medium	2
Low	0

5. Release Recommendation

Recommendation: CONDITIONAL RELEASE

Rationale:

- No Critical defects were found. The core E2E flow (Login → Course Selection → Payment → Course Access) is fully functional.
- DEF_001 (High severity) — Email validation defect must be fixed and verified before production release.
- DEF_002 and DEF_003 (Medium severity) — Can be scheduled for the next sprint patch release.
- The platform is recommended for go-live after DEF_001 fix verification, with DEF_002 and DEF_003 tracked in the next sprint.

6. Sign-Off

Role	Name	Date	Signature
QA Lead	____________________	June 2026	_______________
Product Owner	____________________	June 2026	_______________
Project Manager	____________________	June 2026	_______________